

Practice Problems for Week 5

Operating and Systems Programming 2022/23

1. How does the distinction between kernel mode and user mode function as a rudimentary form of protection (security) system?
2. Which of the following instructions should be privileged?
 - Set value of timer.
 - Read the clock.
 - Clear memory.
 - Issue a trap instruction.
 - Turn off interrupts.
 - Modify entries in device-status table.
 - Switch from user to kernel mode.
 - Access I/O device.
3. Why is it important not to have memory leaks in the kernel?
4. Why is it important that interrupt service routine can be executed in a very small amount of time?
5. What would be the consequences of allowing direct calls to functions of the operating system?